

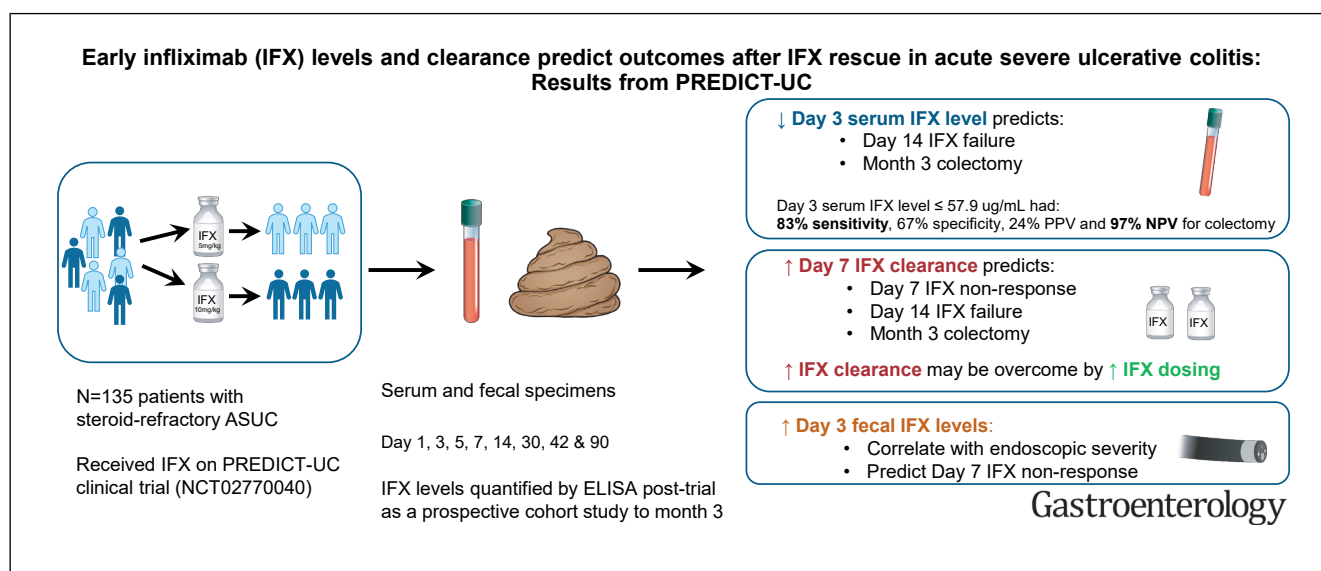
INFLAMMATORY BOWEL DISEASE

Early Infliximab Levels and Clearance Predict Outcomes After Infliximab Rescue in Acute Severe Ulcerative Colitis: Results From PREDICT-UC



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BACKGROUND & AIMS: The role of infliximab therapeutic drug monitoring in acute severe ulcerative colitis (ASUC) management is unknown. We aimed to identify whether infliximab therapeutic drug monitoring is associated with ASUC outcomes. **METHODS:** Serum and stool samples were collected from patients enrolled in the PREDICT-UC

randomized controlled trial (NCT02770040), which compared intensified and standard infliximab rescue in steroid-refractory ASUC. Infliximab levels measured after trial conclusion and clearance derived using pharmacokinetic modelling were correlated with outcomes. **RESULTS:** Infliximab levels were measured in 681 serum and 198 fecal

samples from 135 patients. Lower day 3 serum infliximab levels predicted infliximab failure on day 14 (area under the receiver operator characteristic curve = 0.63; $P = .043$) and colectomy by 3 months (area under the receiver operator characteristic curve = 0.77; $P = .0027$); a threshold of ≤ 57.9 $\mu\text{g/mL}$ had 83% sensitivity, 67% specificity, 24% positive predictive value, and 97% negative predictive value for colectomy. Patients with high clearance between day 1 and 7 (≥ 0.62 L/d) were more likely to respond to an initial 10 mg/kg vs 5 mg/kg infliximab dose (risk ratio, 1.50; 95% confidence interval [CI], 1.01–2.23), and had a higher risk of colectomy if they received an initial 5 mg/kg vs 10 mg/kg dose (HR, 4.81; 95% CI, 1.09–21.37). In patients with high clearance who did not respond to the first infliximab dose, day 14 response rate was higher with a second 10 mg/kg vs 5 mg/kg dose (38% vs 11%; risk ratio, 3.43; 95% CI, 1.05–11.19). Day 3 fecal infliximab levels correlated with endoscopic severity and was associated with day 7 nonresponse ($P = .016$). **CONCLUSIONS:** Early infliximab levels and clearance calculation can predict outcomes in ASUC. This is the first study to demonstrate that high infliximab clearance may be overcome by intensified infliximab dosing. (NCT02770040; Optimizing Infliximab Induction Therapy for Acute Severe Ulcerative Colitis)

Keywords: Therapeutic Drug Monitoring; Biologic; Dose-Response Relationship; Drug; Colectomy.

Acute severe ulcerative colitis (ASUC) is a medical emergency that affects a quarter of patients with ulcerative colitis (UC) at some point in their disease course.¹ Patients who do not respond to first-line treatment with intravenous steroids require rescue therapy with either infliximab or cyclosporin, both of which are considered equally effective in ASUC.^{2,3} Response to rescue therapy is highly variable, with up to 50% of patients not responding to infliximab rescue, thus facing the prospect of sequential rescue therapy or emergency colectomy.⁴ The ability to optimize infliximab use in ASUC may improve response rates and reduce colectomy rates.

Although higher trough infliximab levels have been linked to improved remission rates in moderate-to-severe UC,⁵ the role of therapeutic drug monitoring (TDM) in the ASUC setting is unknown. In particular, it is unclear whether an exposure-response relationship exists for infliximab in ASUC and whether early serum infliximab levels predict outcomes. In addition, there is a paucity of prospective data on factors that determine infliximab clearance in ASUC. Moreover, whether fecal infliximab losses contribute to infliximab clearance and nonresponse is unknown.

We recently performed a multicenter randomized controlled trial (RCT) that examined escalated infliximab dosing strategies in patients with steroid-refractory ASUC.⁶ The PREDICT-UC study did not observe a difference in clinical response to infliximab by day 7 between a 5 mg/kg and a 10 mg/kg initial infliximab dose. However, among patients who received an initial 5 mg/kg dose, response rates were

WHAT YOU NEED TO KNOW

BACKGROUND AND CONTEXT

The role of infliximab therapeutic drug monitoring (TDM) in the management of acute severe ulcerative colitis (ASUC) is not known.

NEW FINDINGS

Low day 3 serum infliximab levels are associated with an increased risk of day 14 infliximab failure and month 3 colectomy; whereas high clearance is associated with poorer outcomes and may be overcome by higher infliximab dosing.

LIMITATIONS

The current infliximab TDM study was observational in nature; clinical management was not adapted in response to infliximab levels.

CLINICAL RESEARCH RELEVANCE

This study supports the use of infliximab TDM in ASUC. Intensive infliximab dosing should be considered in patients with a low day 3 serum infliximab level and those with high clearance or at risk of high clearance. Future studies should explore whether adapting the dosing strategy based on day 3 serum infliximab level improves outcomes.

BASIC RESEARCH RELEVANCE

This study supports the hypothesis that infliximab clearance is a major driver of outcomes in ASUC and that fecal infliximab loss may at least partly contribute to infliximab clearance. Future studies should examine mechanisms contributing to infliximab clearance and non-TNF-mediated pathways in ASUC.

numerically lower in patients with an albumin < 25 g/L (compared with those with an albumin ≥ 25 g/L) and those with a C-reactive protein (CRP) ≥ 50 mg/L (compared with those with a CRP < 50 mg/L). This suggests that patients with a higher inflammatory burden and/or higher clearance may benefit from a higher initial infliximab dose.^{6,7}

The aim of the current study was to examine whether serum or fecal infliximab levels, derived from a prospective cohort of patients, are associated with clinical outcomes in ASUC. We additionally aimed to examine infliximab clearance in ASUC and its relationship to outcomes, as well as factors associated with infliximab clearance in ASUC.

Abbreviations used in this paper: AIS, accelerated induction strategy; ASUC, acute severe ulcerative colitis; AUC, area under the curve; AUROC, area under the receiver operator characteristic curve; CI, confidence interval; CRP, C-reactive protein; ELISA, enzyme-linked immunosorbent assay; HR, hazard ratio; IIS, intensified induction strategy; IQR, interquartile range; NPV, negative predictive value; PK, pharmacokinetic; PPV, positive predictive value; RCT, randomized controlled trial; ROC, receiver operator characteristic; RR, risk ratio; SIS, standard induction strategy; TDM, therapeutic drug monitoring; TNF, tumor necrosis factor; UC, ulcerative colitis; UCEIS, Ulcerative Colitis Endoscopic Index of Severity.

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Materials and Methods

The Clinical PREDICT-UC Study

PREDICT-UC (NCT02770040) was a multicenter RCT across 13 Australian centers that compared intensified dose infliximab strategies with standard dose infliximab in 138 adult patients admitted to hospital with corticosteroid-refractory ASUC.⁶ We enrolled hospitalized patients aged ≥ 18 years with ASUC as per Truelove and Witt's criteria.⁶ Additionally, patients had to be refractory to at least 3 days of intravenous hydrocortisone 100 mg four times a day as assessed using the Oxford criteria. The complete methods have been previously published.⁶

The induction phase consisted of 2 components:

- (1) Randomization 1: RCT of initial infliximab dose (day 0–7). Patients were randomized to receive an initial rescue dose of 10 mg/kg or 5 mg/kg infliximab intravenously (week 0 dose). The primary outcome of clinical response to infliximab was assessed between days 3 and 7.
- (2) Randomization 2: Randomized induction strategy study (day 8 to month 3) comparing intensified (IIS), accelerated (AIS), and standard (SIS) dose induction strategies. The randomization process is explained in more detail in the [Supplementary Material](#).

Patients who did not achieve a clinical response to the induction or salvage strategy prescribed by the clinical trial were withdrawn from the study with subsequent therapy or colectomy determined at treating clinician's discretion.

Clinical data recorded included patient demographics, initial infliximab dose, overall infliximab dosing strategy, and routine biochemical data including CRP and albumin. The Lichtiger score was recorded daily during hospitalization and at each study visit at days 7, 14, 30, 42, and month 3. Biochemical assessments were performed at the same time-points. Patients underwent endoscopic assessment at time of presentation and at month 3 with severity assessed using the Ulcerative Colitis Endoscopic Index of Severity (UCEIS) and Mayo Endoscopic Subscore. Colectomy was assessed to month 3.

As part of the study protocol, serum and stool specimens were collected at day 1, 3, and 5 after the first infliximab dose (or until discharge, whichever occurred earlier). Additionally, samples were collected at day 7, 14, 30, and 42 and month 3 post-first infliximab dose. The current study assessed outcomes to month 3.

Pharmacologic Assessments

The current study was a prospective observational study based on data and specimens derived from patients enrolled in the PREDICT-UC study. Serum and stool specimens were collected and stored at -80°C for subsequent analysis after conclusion of the clinical trial.

Serum specimens were thawed and infliximab level quantified using an enzyme-linked immunosorbent assay (ELISA; MabTrack level infliximab, Essange reagents, the Netherlands) as per the manufacturer's instructions. Samples with high concentrations were further diluted to obtain an absolute infliximab drug concentration. Samples with low infliximab

concentrations ($<2 \mu\text{g/mL}$) were also tested for serum antibody to infliximab (Ridascreen anti-infliximab antibodies ELISA, Darmstadt, Germany).

For fecal infliximab analysis, stool samples were thawed, homogenized, and diluted 1:5 in a solution containing Dulbecco's phosphate-buffered saline (Gibco, ThermoFisher) with 6% bovine serum albumin (Millipore Probumin Bovine Serum Albumin Diagnostic Grade, Merck), using a method that was modified from that previously described by Brandse et al.⁸ In particular, specimens were vortexed briefly, followed by agitation for 30 minutes at 150 rpm on a flatbed rocker. Samples were then centrifuged at 3250g for 10 minutes. The stool supernatant was collected and stored at -80°C for future testing in batches. Fecal infliximab levels were measured using the ELISA described already in this article for serum specimens.

Fecal infliximab levels were only quantified at days 1 and 3 after the initial infliximab dose. The rationale for this was 2-fold. First, a previous study in patients with moderate-to-severe ulcerative colitis found that fecal infliximab concentrations were highest in the first few days after drug administration.⁸ Second, in our study, patients who did not respond to the initial infliximab dose received a salvage infliximab dose at the time of nonresponse between days 3–5. As such, fecal infliximab quantification after day 3 would have been affected by the second dose in these patients.

Because serum and fecal infliximab levels were measured after the conclusion of the clinical trial, they did not influence clinical management or infliximab dosing.

Outcomes

Initial response to infliximab was assessed between days 3 and 7 postinfliximab. Response was defined as a reduction in Lichtiger score to <10 with a decrease of ≥ 3 points from baseline, improvement in rectal bleeding, and decreased stool frequency to ≤ 4 per day. Day 14 infliximab failure was defined as a Lichtiger score ≥ 10 on day 14. Month 3 Mayo endoscopic remission was defined as a Mayo Endoscopic Subscore of ≤ 1 . Combined clinical and endoscopic remission was defined as a partial Mayo Score ≤ 1 and Mayo Endoscopic Subscore ≤ 1 . Month 3 colectomy was defined as colectomies occurring within 3 months of the first infliximab rescue.

Statistical Analysis

Categorical variables were expressed as frequencies and proportions. Skewed continuous variables were expressed as medians with interquartile range (IQR) and compared using the Mann-Whitney *U* test. Relationships between skewed variables were assessed using Spearman rank correlation coefficient. Binary outcome groups included: response by day 7, day 14 infliximab failure, and colectomy by 3 months. Serum and fecal infliximab levels at days 1 and 3 were analyzed for their association with key outcomes. Levels beyond day 3 were not directly compared because initial infliximab nonresponders received further infliximab doses between day 4 and 7, which would have confounded the results.

The discriminative utility of candidate predictors was assessed using nonparametric receiver operator characteristic (ROC) curve analysis and by calculating the area under the receiver operator characteristic curve (AUROC). Where

predictors were negatively correlated with outcomes, AUROC was calculated using the inverse of the value of the predictor. Optimal thresholds for candidate predictors were determined using Youden index. The performance of dichotomized predictors was further assessed by reporting the sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) for each outcome.

Parametric estimation of infliximab clearance and exposure (influximab area under the curve [AUC]) was performed using a 2-compartment pharmacokinetic (PK) model with first order elimination. Values for the central volume of distribution (V_1), peripheral volume of distribution (V_2), and inter-compartmental clearance (Q) parameters were fixed according to previously estimated formulas and values.^{9,10} Specifically, we set $V_1 = 3.29 \times ([\text{weight, kg}]/77)^{0.538} \times 0.863^{[\text{female}]}$, $V_2 = 4.13$, $Q = 0.1668$. Individual clearance (Cl) was then estimated using maximum likelihood estimation, assuming a log-normal distribution for measured infliximab concentrations. Clearance was estimated at day 3 using serum infliximab levels from day 1 and day 3; at day 7 using levels from day 5 and day 7; at day 30 using levels from day 14 and day 30; and at day 90 using levels from day 42 and day 90. Early serum infliximab clearance was defined as clearance based on serum infliximab levels from day 1–7. Undetectable concentrations were given a value at half the lower limit of quantification. Infliximab AUC was estimated parametrically using the fitted PK profile after estimating individual clearance. We did not calculate infliximab AUC beyond day 3 for the outcomes of day 7 response, day 14 failure, and colectomy as nonresponders to the first infliximab dose received a second dose as early as day 3, which would have skewed the results. Nonparametric method estimation of infliximab AUC was also performed by using the trapezoidal rule for observed infliximab concentrations. Parametric infliximab AUCs at days 3, 7, 30, and 90 were calculated to assess their association with month 3 remission.

A subgroup analysis comparing an initial infliximab dose of 5 mg/kg vs 10 mg/kg was performed by dividing the cohort into 2 halves according to estimated clearance based on the median clearance. Risk ratios (RR) and confidence intervals (CI) were estimated using modified Poisson regression with cluster-robust standard errors to account for center-based clustering. Hazard ratios (HRs) for colectomy were estimated using Cox proportional hazards regression. Patients entered the risk set on the day of infliximab administration (designated as day 0) and were followed up until either colectomy or censoring. Patients were censored if they were lost to follow-up, withdrew consent, or completed the 3-month follow-up period, whichever came first. Robust standard errors were used to account for center-based clustering. The original randomizations from the PREDICT-UC RCT were included as an indicator covariate in all regression models. A P value of $<.05$ was considered statistically significant. No adjustments for multiple comparisons were made due to the exploratory nature of the study. The R (version 4.3.0) statistical programming language in RStudio (version 2023.06.1+524) was used for statistical analysis.

Ethical Considerations

This study was approved by the Austin Health Human Research Ethics Committee and all participating centers via a high-risk national ethics application (HREC/14/Austin/595).

Written informed consent was obtained from all study participants.

Results

Participant Characteristics and Outcomes

Of the 138 patients who received infliximab on the PREDICT-UC trial, 135 had at least 1 serum or fecal infliximab level quantified. Of the 135, 44 received an initial intensified 10 mg/kg infliximab dose whereas 91 received an initial standard 5 mg/kg infliximab dose. Outcomes are shown in [Supplementary Figure 1](#) and baseline characteristics of the 135 patients included in the current infliximab study are summarized in [Table 1](#).

Of the 135 patients, 85 (63%) demonstrated initial response by day 7, 37 (27%) experienced day 14 infliximab failure, and 17 (13%) required colectomy by month 3. Eighty-two patients completed the study to month 3, of which 81 had endoscopic assessment at that timepoint. Of the 81 patients, 63 (78%) were in Mayo endoscopic remission at month 3 and 57 (70%) achieved combined clinical and endoscopic remission at the same timepoint.

Serum Infliximab Levels

Infliximab levels were quantified in 681 serum specimens collected across 134 patients. Median serum infliximab level at day 1 postinfliximab was 175.4 $\mu\text{g/mL}$ (IQR, 137.2–202.7) in patients who received an initial 10 mg/kg dose compared with 91.8 $\mu\text{g/mL}$ (IQR, 77.2–109.4) in those who received an initial 5 mg/kg dose ($P < .0001$). Similarly, at day 3 postinfliximab, patients who received a 10 mg/kg initial dose had a higher serum infliximab level compared with those who received a 5 mg/kg dose (116.3 $\mu\text{g/mL}$; IQR, 83.4–132.9 vs 56.0 $\mu\text{g/mL}$; IQR, 46.0–67.0; $P < .0001$).

Among patients who completed the study to month 3, the day 90 serum infliximab AUC was similar in the 3 induction groups: IIS (2982 $\mu\text{g}\cdot\text{d/mL}$; IQR, 2022–3947), AIS (2909 $\mu\text{g}\cdot\text{d/mL}$; IQR, 2127–4032), and SIS (3170 $\mu\text{g}\cdot\text{d/mL}$; IQR, 2233–3774). Day 90 serum infliximab levels were highest in the SIS group (6.70 $\mu\text{g/mL}$; IQR, 4.20–9.65), followed by the AIS (3.95 $\mu\text{g/mL}$; IQR, 2.22–7.40), and lowest in the IIS group (1.60 $\mu\text{g/mL}$; IQR, 0.75–3.55; $P = .00162$).

Initial infliximab response by day 7. Day 3 serum infliximab levels was 69.0 $\mu\text{g/mL}$ (IQR, 56.2–98.9) in responders compared with 57.4 $\mu\text{g/mL}$ (IQR, 48.2–76.3) in nonresponders although this did not reach statistical significance ($P = .087$; [Table 2](#)). Among patients who received an initial 5 mg/kg infliximab dose, however, day 7 infliximab responders had a significantly higher serum infliximab level at day 3 (61.8 $\mu\text{g/mL}$; IQR, 47.9–69.0) compared with nonresponders (50.6 $\mu\text{g/mL}$; IQR, 45.0–57.6; $P = .017$; [Figure 1A](#)). This pattern was, however, not observed in patients who received an initial 10 mg/kg dose ([Figure 1D](#)).

Infliximab failure at day 14. Day 3 serum infliximab level was 55.4 $\mu\text{g/mL}$ (IQR, 45.4–76.0) in patients who experienced infliximab failure at day 14 compared with 69.0 $\mu\text{g/mL}$ (IQR, 55.7–101.3) in day 14 infliximab

Table 1. Baseline Demographics of the 135 Patients With Available Serum Infliximab Levels

Characteristic	n (%)
Age (y), median (IQR)	33 (24–42)
Male sex	73 (54%)
Weight (kg)	74.6 ± 17.8
Disease duration, y, median (IQR) (n = 95)	2.8 (0.8–10.0)
Smoking status (n = 135)	
Nonsmoker	76 (56%)
Ex-smoker	54 (40%)
Active smoker	5 (4%)
Known disease extent at study entry (Montreal classification) (n = 90)	
E1	14 (16%)
E2	40 (44%)
E3	36 (40%)
Baseline medications	
Aminosalicylates	66/135 (49%)
Steroids	96/135 (71%)
Thiopurine	26/135 (19%)
Methotrexate	3/135 (2%)
Anti-TNF	0 (0%)
Anti-integrin	0 (0%)
Calcineurin inhibitor	1/135 (1%)
Prior medications	
Aminosalicylates	23/135 (17%)
Steroids	19/135 (14%)
Thiopurine	20/135 (15%)
Methotrexate	4/135 (3%)
Anti-TNF	7/135 (5%)
Anti-integrin	2/135 (1%)
Calcineurin inhibitor	2/135 (1%)
Duration of flare prior to hospitalization (wk), median (IQR) (n = 129)	4 (2–7)
Previous episode of ASUC (n = 94)	30 (32%)
Baseline Mayo Endoscopic Subscore, median (IQR)	3 (2–3)
Baseline UCEIS (IQR)	6 (5–7)
Montreal extent at index endoscopy	
E1	7/135 (5%)
≥E2	128/135 (95%)
Screening Lichtiger score, median (IQR) (n = 104)	13 (11–15)
Screening stool frequency/24 h, median (IQR) (n = 101)	10 (7–13)
Admission hematology & biochemistry	
Hb (g/L)	128.2 ± 19.9
White cell count (×10 ⁹ /L)	11.0 (8.7–13.8)
CRP (mg/L) (n = 131)	49.7 (26.4–109.3)
Serum albumin (g/L) (n = 127)	31.8 ± 6.4
Day 0 biochemistry (day of infliximab administration)	
CRP (n = 133)	20.6 (5.7–44.0)
Albumin (n = 131)	27.1 ± 5.4

Table 1. Continued

Characteristic	n (%)
Initial infliximab dose	
5 mg/kg	91 (67%)
10 mg/kg	44 (33%)

NOTE. Categorical variables are expressed as n (%). Continuous variables are expressed as median (IQR) or as mean (SD). Denominators are n = 135, unless otherwise stated. “n” denotes patients with available data. Hb, hemoglobin.

responders (AUROC, 0.63; 95% CI, 0.50–0.75; $P = .043$; [Table 3](#)). Comparable results were observed on subgroup analysis of the initial 5 mg/kg dosing group ([Figure 1B](#)). In the smaller 10 mg/kg dosing group, a similar trend was observed but statistical significance was not reached ([Figure 1E](#)). A day 3 serum infliximab level $\leq 53.6 \mu\text{g/mL}$ had a 50% sensitivity, 75.6% specificity, 42.4% PPV, and 80.8% NPV for infliximab failure at day 14.

Colectomy. A higher day 3 serum infliximab was associated with a lower risk of colectomy by 3 months. In patients who avoided colectomy, day 3 serum infliximab was $69.0 \mu\text{g/mL}$ (IQR, 51.1–101.7) compared with $51.0 \mu\text{g/mL}$ (IQR, 39.2–57.4) in those who required colectomy (AUROC, 0.77; 95% CI, 0.64–0.89; $P = .0027$; [Table 4](#)). Similar observations were noted in subgroup analyses of the initial 5 mg/kg and 10 mg/kg infliximab dosing groups, although statistical significance was not reached in the latter group ([Figure 1C](#) and [Figure 1F](#)). Using Youden index, a day 3 serum infliximab level $\leq 57.9 \mu\text{g/mL}$ predicted colectomy with 83.3% sensitivity, 67% specificity, 24.4% PPV, and 96.9% NPV.

Month 3 remission. In patients who completed the study to month 3, early serum infliximab levels did not predict Mayo endoscopic remission at month 3. However, serum infliximab levels at month 3 (day 90) were higher in patients who achieved Mayo endoscopic remission at month 3 compared with those with active endoscopic disease ($4.2 \mu\text{g/mL}$; IQR, 2.0–8.9 vs $2.0 \mu\text{g/mL}$; IQR, 0.9–4.3; AUROC, 0.69; 95% CI, 0.54–0.85; $P = .017$; [Supplementary Table 1](#)). Using Youden index, a day 90 serum infliximab level $\geq 2.50 \mu\text{g/mL}$ predicted Mayo endoscopic remission at month 3 with 71.7% sensitivity, 68.8% specificity, 88.4% PPV, and 42.3% NPV.

Antibodies to Infliximab

Antibodies to infliximab were tested in 25 samples across 23 patients with low serum infliximab levels ($<2 \mu\text{g/mL}$). Antibodies to infliximab levels were undetectable ($<2.5 \text{ ng/mL}$) in all specimens, except for 1 patient who developed high antibody levels ($>124.99 \text{ ng/mL}$) at both day 30 and day 42 postinfliximab. This patient had previous exposure to infliximab (last dose 3 years before study entry) and was randomized to intensified infliximab rescue on the study. The patient responded to the initial 10 mg/kg at week 0 and received a second dose of 10 mg/kg at week 1 as per protocol and went on to receive azathioprine maintenance therapy.

Table 2. Relationship Between Infliximab Parameters and Initial Infliximab Response by Day 7

Variable	n	Day 7 response (n = 85)	Day 7 nonresponse (n = 50)	P
Initial 5 mg/kg group				
Day 1 serum IFX level ($\mu\text{g/mL}$)	78	93.8 (77.2–108.7)	88.7 (77.7–110.4)	.75
Day 3 serum IFX level ($\mu\text{g/mL}$)	68	61.8 (47.9–69.0)	50.6 (45.0–57.6)	.017
Nonparametric IFX day 3 AUC (mg)	59	265.0 (231.4–296.1)	247.8 (223.9–292.0)	.52
Parametric (estimated) IFX day 3 AUC (mg)	88	267.9 (230.1–294.3)	243.2 (221.0–275.3)	.12
Day 1 fecal IFX level ($\mu\text{g/mL}$)	63	1.1 (0.1–4.7)	4.9 (0.6–10.1)	.042
Day 3 fecal IFX level ($\mu\text{g/mL}$)	67	1.1 (0.1–2.4)	3.1 (1.1–7.6)	.015
Initial 10 mg/kg group				
Day 1 serum IFX level ($\mu\text{g/mL}$)	37	166.4 (132.6–183.4)	191.6 (175.9–212.9)	.083
Day 3 serum IFX level ($\mu\text{g/mL}$)	38	116.3 (95.1–132.4)	117.7 (77.1–132.1)	.52
Nonparametric IFX day 3 AUC (mg)	31	489.3 (416.0–569.5)	533.9 (446.7–577.0)	.65
Parametric (estimated) IFX day 3 AUC (mg)	44	504.9 (441.4–573.7)	555.7 (455.1–574.6)	.59
Day 1 fecal IFX level ($\mu\text{g/mL}$)	35	9.4 (2.2–12.1)	3.8 (1.3–23.6)	.80
Day 3 fecal IFX level ($\mu\text{g/mL}$)	33	3.6 (1.4–7.3)	5.7 (4.5–7.8)	.31
All patients				
Day 1 serum IFX level ($\mu\text{g/mL}$)	115	108.4 (84.5–137.7)	104.0 (81.7–128.5)	.76
Day 3 serum IFX level ($\mu\text{g/mL}$)	106	69.0 (56.2–98.9)	57.4 (48.2–76.3)	.087
Nonparametric IFX day 3 AUC (mg)	90	300.0 (241.5–419.2)	292.0 (241.1–399.9)	.58
Parametric (estimated) IFX day 3 AUC (mg)	132	294.7 (244.8–443.0)	273.6 (233.1–406.5)	.28
Estimated day 3 IFX clearance (L/d)	131	0.70 (0.51–0.95)	0.73 (0.61–0.98)	.44
Estimated day 7 IFX clearance (L/d)	116	0.51 (0.38–0.66)	0.76 (0.54–0.90)	<.0001
Change in IFX clearance from day 3–7 (%)	115	–29.3 (–45.1 to –9.9)	–17.4 (–24.1 to 11.5)	<.0001 ^a
Estimated overall early IFX clearance (L/d)	132	0.56 (0.39–0.72)	0.72 (0.55–0.89)	.00012
Day 1 fecal IFX level ($\mu\text{g/mL}$)	98	2.4 (0.2–9.4)	4.1 (0.6–10.7)	.33
Day 3 fecal IFX level ($\mu\text{g/mL}$)	100	1.4 (0.2–5.1)	4.7 (1.6–7.8)	.016

AUC, area under the curve; IFX, infliximab; n, number of patients with available data.

^aP value derived using analysis of covariance.

Fecal Infliximab Levels

Day 1 fecal infliximab levels were higher in patients who received an initial 10 mg/kg infliximab dose (median, 7.2 $\mu\text{g/mL}$; IQR, 1.8–12.8) compared with those who received a 5 mg/kg dose (1.5 $\mu\text{g/mL}$; IQR, 0.2–7.0; $P = .0093$). Similarly, day 3 fecal infliximab level was 4.7 $\mu\text{g/mL}$ (IQR, 1.6–7.6) in the 10 mg/kg group compared with 1.5 $\mu\text{g/mL}$ (IQR, 0.2–5.8) in the 5 mg/kg group ($P = .0063$). Fecal infliximab levels correlated positively with CRP, endoscopic severity as assessed using UCEIS, and UCEIS subscores, including bleeding and the presence of erosions/ulcers (Supplementary Table 2).

A lower day 3 fecal infliximab level was associated with day 7 response (1.4 $\mu\text{g/mL}$; IQR, 0.2–5.1 in responders vs 4.7 $\mu\text{g/mL}$; IQR, 1.6–7.8 in nonresponders; $P = 0.016$). On subgroup analysis, this was observed in the 5 mg/kg group but not in the 10 mg/kg, possibly due to saturation of fecal losses in the latter group (Table 2; Figure 2A and 2D). Day 1 and 3 fecal infliximab levels were also numerically higher in patients who failed infliximab at day 14 and those who required colectomy by month 3, although these did not reach statistical significance (Table 3, Table 4). Subgroup analyses by initial infliximab dosing groups are presented in Figure 2.

A low fecal infliximab at day 1 predicted both Mayo remission at month 3 (AUROC, 0.79; 95% CI: 0.65–0.92) and combined clinical and endoscopic Mayo remission at

month 3 (AUROC, 0.69; 95% CI, 0.54–0.84). In patients who went on to achieve Mayo endoscopic remission at month 3, median fecal infliximab at day 1 was 0.6 $\mu\text{g/mL}$ (IQR, 0.1–4.1) compared with 10.2 $\mu\text{g/mL}$ (IQR, 4.3–14.0) in those with active endoscopic disease at month 3 ($P = .0011$; Supplementary Table 1). Using Youden index, a day 1 fecal infliximab level of ≤ 2.1 $\mu\text{g/mL}$ had 67.4% sensitivity, 86.7% specificity, 93.5% PPV, and 48.1% NPV for Mayo endoscopic remission at month 3. Similarly, median day 1 fecal infliximab level was 1.1 $\mu\text{g/mL}$ (IQR, 0.1–4.6) in patients who would go on to achieve combined clinical and endoscopic remission at month 3 compared with 6.6 $\mu\text{g/mL}$ (IQR, 0.9–12.8) in those with active disease ($P = .021$; Supplementary Table 3).

Infliximab Exposure (AUC)

Early AUC (day 3 infliximab AUC) was not associated with infliximab response by day 7 or day 14 (Table 2, Table 3). However, nonparametric day 3 infliximab AUC was higher in patients who avoided colectomy compared with those who required colectomy (302.1 mg; IQR, 243.0–440.5 vs 247.8 mg; IQR, 228.7–280.8; $P = .034$; Table 4). Patients who were in endoscopic remission at month 3 also had a higher serum infliximab AUC between day 1 and day 90 of 3181.2 $\mu\text{g}\cdot\text{d/mL}$ (IQR, 2210.0–4219.8) vs 2788.1 $\mu\text{g}\cdot\text{d/mL}$ (IQR, 1771.8–3204.1) in those with endoscopically active disease ($P = .046$; Supplementary Table 1).

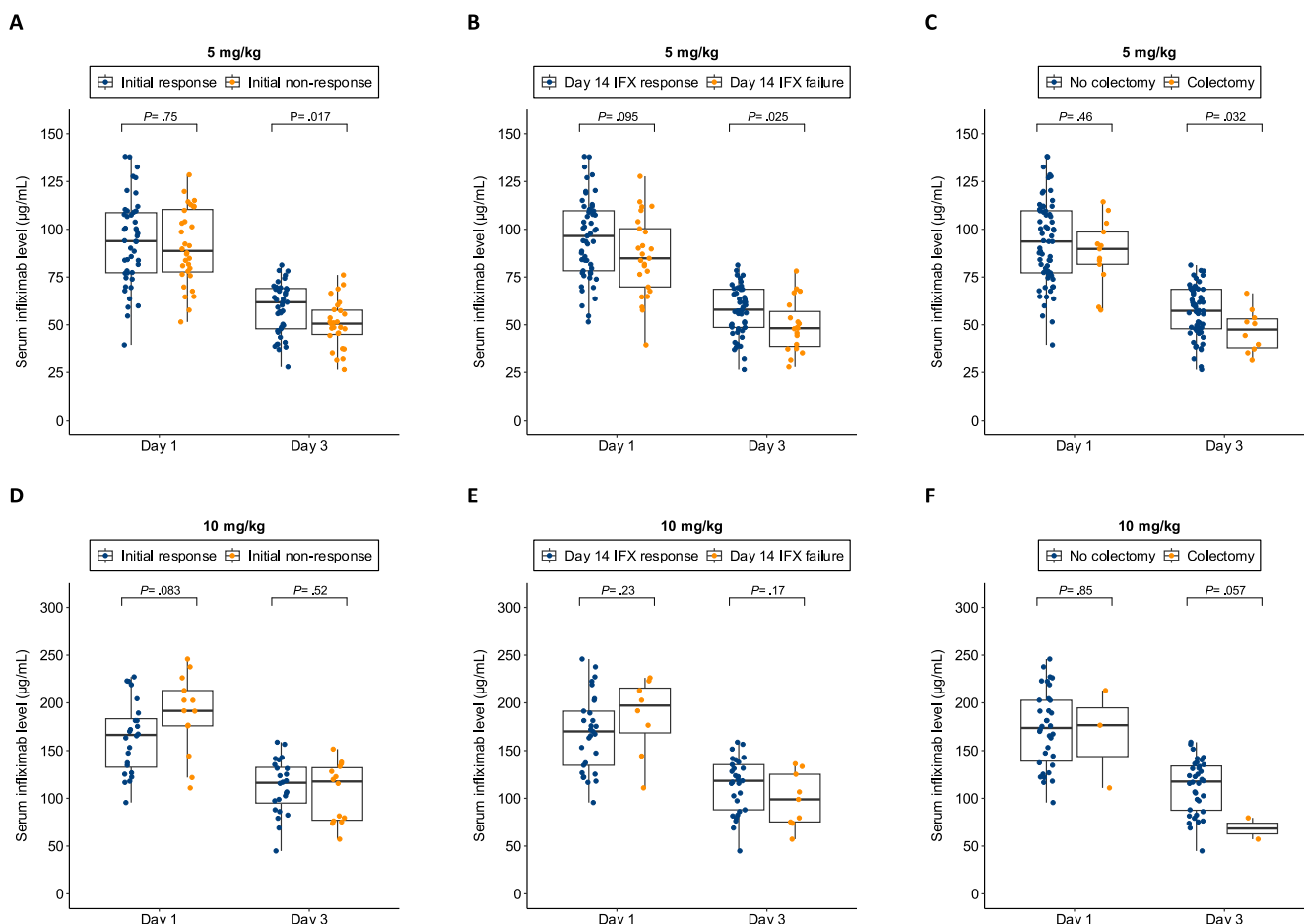


Figure 1. Relationship between serum IFX level and initial IFX response by day 7, IFX failure at day 14, and colectomy by month 3 in patients who received an initial 5 mg/kg IFX dose (A, B, and C, respectively) and those who received an initial 10 mg/kg IFX dose (D, E, and F, respectively).

Infliximab Clearance

Infliximab clearance was highest in the first 3 days after infliximab administration and gradually decreased over time to day 30 (Figure 3A). Median infliximab clearance at day 3 was 0.72 L/day (IQR, 0.54–0.98) compared with 0.56 L/day (IQR, 0.41–0.76) at day 7 ($P < .0001$). Day 7 clearance was in turn higher than day 30 clearance (median, 0.36 L/d; IQR, 0.28–0.49; $P < .0001$). Day 30 clearance did not differ from day 90 clearance (median, 0.39 L/d; IQR, 0.32–0.56; $P = .085$).

Association between infliximab clearance, fecal losses, and pharmacotherapy. Overall early infliximab clearance (within 1 week of initial infliximab dose) was higher in patients receiving an initial dose of 10 mg/kg compared with 5 mg/kg (median, 0.69 L/d [IQR, 0.56–0.85] vs 0.57 [IQR, 0.39–0.74]; $P = .0037$). Early clearance weakly correlated with both day 1 stool infliximab ($\rho = .36$; $P = .00027$) and day 3 stool infliximab ($\rho = .36$; $P = .00029$). Early clearance was not different in patients on baseline thiopurine compared with patients not on thiopurine at baseline (0.51 L/d [IQR, 0.38–0.75] vs 0.66 [IQR, 0.49–0.81]; $P = .16$).

Association between infliximab clearance and conventional factors. Early clearance was higher in men

(median, 0.73 L/d; IQR, 0.57–0.85) compared with women (median, 0.49 L/d; IQR, 0.37–0.63; $P < .0001$), and was also correlated with weight ($\rho = .44$; $P < .0001$). Early clearance moderately correlated with day 0 CRP ($\rho = .43$; $P < .0001$), baseline UCEIS ($\rho = .31$; $P = .0030$), and negatively correlated with day 0 albumin ($\rho = -.27$; $P = .0020$), but not day 0 stool frequency ($\rho = .01$; $P = .91$).

Relationship between infliximab clearance and outcomes. Day 3 estimated infliximab clearance did not differ between day 7 infliximab responders and non-responders, nor did it differ between patients who required colectomy and those who avoided colectomy at month 3 (Figures 3B and 3D; Tables 2 and Table 4). However, day 3 infliximab clearance was higher in patients who failed infliximab at day 14 compared with those who demonstrated a response (0.87 L/d [IQR, 0.69–1.10] vs 0.67 L/d [IQR, 0.51–0.91]; $P = .011$; Table 3; Figure 3C).

Day 7 infliximab clearance was higher in those who demonstrated an initial nonresponse by day 7 compared with responders (median, 0.76 L/d [IQR, 0.54–0.90] vs 0.51 L/d [IQR, 0.38–0.66]; $P < .0001$; Table 2; Figure 3B). Similarly, day 7 infliximab clearance was higher in those who failed infliximab at day 14 compared with responders (median, 0.79 L/d [IQR, 0.66–0.92] vs 0.51 L/d [IQR, 0.38–

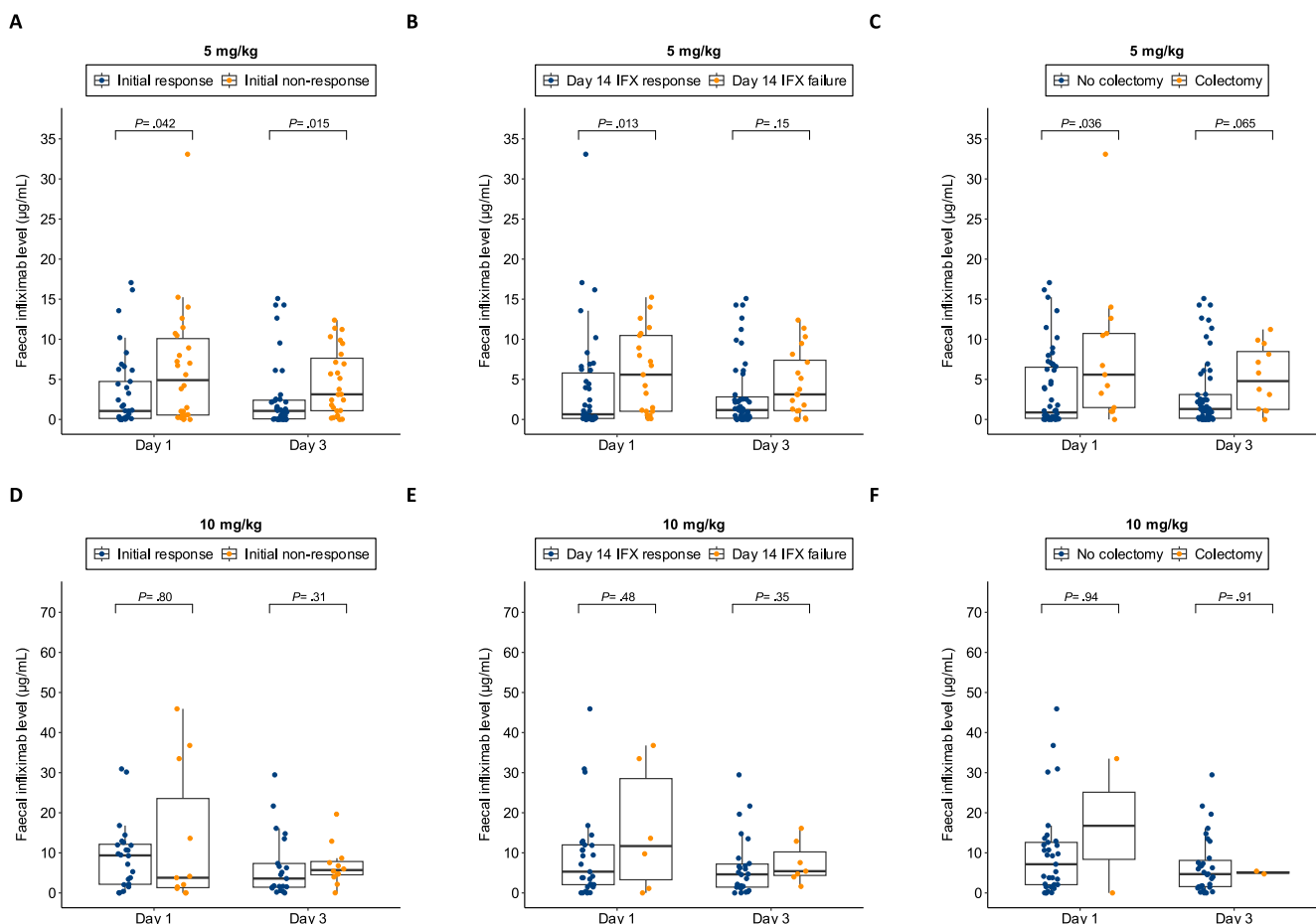


Figure 2. Relationship between faecal IFX level and initial IFX response by day 7, IFX failure at day 14, and colectomy by month 3 in patients who received an initial 5 mg/kg IFX dose (A, B, and C, respectively) and those who received an initial 10 mg/kg IFX dose (D, E, and F, respectively).

0.67]; $P < .0001$; Figure 3C; Table 3), and those who required colectomy by 3 months compared with those who avoided colectomy (median, 0.76 L/d [IQR, 0.67–0.91] vs 0.53 L/d [IQR, 0.39–0.71] $P = .0007$; Figure 3D; Table 4).

When considering the dynamic change in clearance over time, there was a greater reduction in infliximab clearance from day 3 to day 7 in day 7 responders compared with nonresponders [−29.3% [IQR, −45.1 to −9.9%] vs −17.4% [IQR, −24.1 to 11.5%]; $P < .0001$; Table 2). There was also a greater reduction in clearance from day 3 to 7 in patients who avoided colectomy compared with those who required colectomy by month 3 [−24.1% [IQR, −40.9 to −5.3%] vs −19.1% [IQR, −22.8 to 23.8%]; $P = .0053$; Table 4).

The overall early infliximab clearance based on serum infliximab levels between days 1 and 7 was also significantly associated with and predicted outcomes including day 7 infliximab nonresponse (AUROC, 0.70; 95% CI, 0.60–0.80), day 14 infliximab failure (AUROC, 0.77; 95% CI, 0.67–0.87), and month 3 colectomy (AUROC, 0.69; 95% CI, 0.55–0.83; Tables 2, 3, and 4). Although early infliximab clearance did not predict Mayo endoscopic remission at month 3, reduction in infliximab clearance from day 3 to 7 was significantly higher in patients who went on to achieve remission at month 3 compared with those who did not

[−29.3% [IQR, −48.0 to −14.3%] vs −10.1% [IQR, −26.5 to 3.9%], $P = .020$; Supplementary Table 1). The addition of early clearance to day 0 CRP (log transformed) significantly improved a Poisson model for initial day 7 nonresponse ($P = .029$) as well as a model for day 14 infliximab failure ($P = .0023$), but not a Cox model for colectomy ($P = .52$) using the likelihood ratio test for nested models. The respective discriminative utility (AUROC) for bivariable models (infliximab clearance and log CRP) predicting initial day 7 nonresponse and day 14 infliximab failure were 0.72 (95% CI, 0.62–0.81) and 0.78 (95% CI, 0.68–0.87). For comparison, the discriminative utility (AUROC) of baseline (day 0) CRP alone in predicting day 7 nonresponse, day 14 infliximab failure, and colectomy were 0.64 (95% CI, 0.54–0.73), 0.65 (95% CI, 0.55–0.75), and 0.72 (95% CI, 0.61–0.84), respectively.

On multivariable Poisson regression, independent predictors of initial day 7 nonresponse included overall early infliximab clearance (RR, 3.96 per L/d; 95% CI, 1.83–8.58; $P = .0005$), a higher Lichtiger score on day of infliximab administration (RR, 1.08 per point; 95% CI, 1.03–1.14; $P = .0038$), and baseline thiopurine use (RR, 0.54; 95% CI, 0.35–0.82; $P = .0043$; Supplementary Table 4). Factors independently associated with day 14 infliximab failure

Table 3. Relationship Between Infliximab Parameters and Infliximab Failure at Day 14

Variable	n	Day 14 IFX response (n = 98)	Day 14 IFX failure (n = 37)	P
Initial 5 mg/kg group				
Day 1 serum IFX level ($\mu\text{g/mL}$)	78	96.5 (78.3–109.7)	84.8 (69.8–100.3)	.095
Day 3 serum IFX level ($\mu\text{g/mL}$)	68	57.9 (48.6–68.6)	48.2 (38.7–57.0)	.025
Initial 10 mg/kg group				
Day 1 serum IFX level ($\mu\text{g/mL}$)	37	170.1 (134.6–191.3)	197.1 (168.5–215.4)	.23
Day 3 serum IFX level ($\mu\text{g/mL}$)	38	118.4 (88.0–135.3)	98.9 (75.3–125.2)	.17
All patients				
Day 1 serum IFX level ($\mu\text{g/mL}$)	115	110.1 (87.8–138.0)	91.4 (78.1–114.4)	.093
Day 3 serum IFX level ($\mu\text{g/mL}$)	106	69.0 (55.7–101.3)	55.4 (45.4–76.0)	.043
Nonparametric IFX day 3 AUC (mg)	90	301.1 (242.2–433.7)	277.6 (231.1–345.8)	.20
Parametric (estimated) IFX day 3 AUC (mg)	132	294.9 (248.9–443.0)	260.3 (220.1–341.9)	.058
Estimated day 3 IFX clearance (L/d)	131	0.67 (0.51–0.91)	0.87 (0.69–1.10)	.011
Estimated day 7 IFX clearance (L/d)	116	0.51 (0.38–0.67)	0.79 (0.66–0.92)	<.0001
Change in IFX clearance from day 3–7 (%)	115	–24.8 (–41.0 to –7.1)	–17.4 (–27.4 to 11.4)	.065 ^a
Estimated overall early IFX clearance (L/d)	132	0.56 (0.39–0.71)	0.76 (0.70–0.92)	<.0001
Day 1 fecal IFX level ($\mu\text{g/mL}$)	98	2.2 (0.2–8.8)	6.7 (1.1–11.1)	.092
Day 3 fecal IFX level ($\mu\text{g/mL}$)	100	1.6 (0.4–6.1)	4.0 (1.5–7.8)	.12

Abbreviations as in Table 2.

^aP value derived using analysis of covariance.

included early infliximab clearance (RR, 6.89 per L/d; 95% CI, 2.35–20.16; $P = 0.0004$), day 0 CRP (RR, 1.21 per log mg/L; 95% CI, 1.06–1.38; $P = .0048$), and standard infliximab induction strategy (RR, 1.73; 95% CI, 1.12–2.68; $P = .013$; [Supplementary Table 5](#)). For the outcome of colectomy, only a higher CRP at day 0 was independently predictive in a multivariable Cox regression model (RR, 2.05 per log mg/L; 95% CI, 1.17–3.60; $P = .012$; [Supplementary Table 6](#)). Endoscopic severity assessed using the UCEIS or Mayo endoscopic score did not independently predict day 7 nonresponse, day 14 infliximab failure, or month 3 colectomy.

Estimated day 90 infliximab clearance was higher in patients who had active endoscopic disease at month 3 compared with those in remission (0.46 L/d [IQR, 0.34–0.79] vs 0.37 L/d [IQR, 0.30–0.50]; $P = .043$; [Supplementary Table 1](#)).

Escalated Infliximab Dosing Stratified by Clearance

In a post-hoc subgroup analysis, patients with high early clearance ($\text{CI} \geq 0.62$ L/d) were less likely than patients with low clearance ($\text{CI} < 0.62$ L/d) to demonstrate an initial infliximab response by day 7 (31/66 [47%] vs 52/66 [79%]; RR, 0.60; 95% CI, 0.45–0.79; $P = .00023$), more likely to fail infliximab at day 14 (29/66 [44%] vs 7/66 [11%]; RR, 4.14; 95% CI, 2.03–8.44; $P < .0001$), and require colectomy by 3 months (15/65 [23%] vs 2/65 [3%]; HR, 8.18; 95% CI, 1.87–35.77; $P = .0053$).

Patients with high clearance were more likely to demonstrate an initial infliximab response to a 10 mg/kg dose compared with 5 mg/kg (16/27 [59%] vs 15/39 [38%]; RR, 1.50; 95% CI, 1.01–2.23; $P = .046$) after accounting for thiopurine exposure as a stratification variable. There was no difference in response to 10 mg/kg vs

5 mg/kg in patients with low early clearance (13/17 [76%] vs 39/49 [80%]; RR, 0.94; 95% CI, 0.74–1.19; $P = .60$). Patients with high early clearance were more likely to require colectomy if they were allocated to the AIS/SIS arm compared with the IIS arm (13/39 [33%] vs 2/26 [8%]; HR, 4.81; 95% CI, 1.09–21.37; $P = .039$). There was no significant difference in colectomy rate between initial 5 mg/kg vs 10 mg/kg in low early clearance patients (1/48 [2%] vs 1/17 [6%]; HR, 0.34; 95% CI, 0.02–5.41; $P = .44$). On multivariable regression across the whole cohort, a standard infliximab induction strategy was also associated with an increased risk of day 14 infliximab failure compared with the intensified induction strategy (RR, 1.75; 95% CI, 1.15–2.64; $P = .0083$; [Supplementary Table 5](#)).

Patients with high early clearance were more likely to fail infliximab by 14 days if they received standard induction compared with intensified/accelerated induction (11/19 [58%] vs 18/47 [38%]; RR, 1.42; 95% CI, 1.04–1.92; $P = .027$). There was no significant difference in day 14 failure rates between standard and intensified/accelerated induction in patients with low early clearance (4/24 [17%] vs 3/42 [7%]; RR, 2.33; 95% CI, 0.68–7.97; $P = .18$). In patients with high early clearance who did not respond to the initial infliximab dose, response by day 14 was higher in patients who received a second 10 mg/kg dose compared with 5 mg/kg (10/26 [38%] vs 1/9 [11%]; RR, 3.43; 95% CI, 1.05–11.19; $P = .041$). There was no significant difference between a second dose of 10 mg/kg and 5 mg/kg in initial nonresponders who had low early clearance (7/8 [88%] vs 3/6 [50%]; RR, 1.56; 95% CI, 0.85–2.87; $P = .15$).

Discussion

This large prospective observational study provides insights into the underlying pharmacokinetic drivers of

Table 4. Relationship Between Infliximab Parameters and Colectomy by Month 3

Variable	n	No colectomy (n = 118)	Colectomy (n = 17)	P
Initial 5 mg/kg group				
Day 1 serum IFX level ($\mu\text{g/mL}$)	78	93.6 (77.2–109.7)	89.7 (81.7–98.6)	.46
Day 3 serum IFX level ($\mu\text{g/mL}$)	68	57.3 (47.9–68.6)	47.5 (38.0–53.1)	.032
Initial 10 mg/kg group				
Day 1 serum IFX level ($\mu\text{g/mL}$)	37	173.7 (139.0–202.7)	176.6 (143.8–194.8)	.85
Day 3 serum IFX level ($\mu\text{g/mL}$)	38	117.7 (87.5–133.9)	68.4 (62.8–74.0)	.057
All patients				
Day 1 serum IFX level ($\mu\text{g/mL}$)	115	109.7 (83.9–141.2)	91.8 (83.2–110.2)	.12
Day 3 serum IFX level ($\mu\text{g/mL}$)	106	69.0 (51.1–101.7)	51.0 (39.2–57.4)	.0027
Nonparametric IFX day 3 AUC (<i>mg</i>)	90	302.1 (243.0–440.5)	247.8 (228.7–280.8)	.034
Parametric (estimated) IFX day 3 AUC (<i>mg</i>)	132	294.7 (243.2–448.9)	250.6 (223.7–310.4)	.082
Estimated day 3 IFX clearance (<i>L/d</i>)	131	0.71 (0.52–0.97)	0.84 (0.69–1.01)	.38
Estimated day 7 IFX clearance (<i>L/d</i>)	116	0.53 (0.39–0.71)	0.76 (0.67–0.91)	.0007
Change in IFX clearance from day 3–7 (%)	115	–24.1 (–40.9 to –5.3)	–19.1 (–22.8 to 23.8)	.0053 ^a
Estimated overall early IFX clearance (<i>L/d</i>)	132	0.58 (0.41–0.76)	0.76 (0.70–0.85)	.011
Day 1 fecal IFX level ($\mu\text{g/mL}$)	98	2.4 (0.2–9.4)	5.6 (1.2–11.7)	.29
Day 3 fecal IFX level ($\mu\text{g/mL}$)	100	2.0 (0.4–6.2)	5.1 (1.7–7.9)	.18

Abbreviations as in Table 2.

^aP value derived using analysis of covariance.

infliximab response and failure in ASUC. Importantly, we found that early infliximab levels and clearance predict outcomes in ASUC. Additionally, we are the first to demonstrate that a high early infliximab clearance can be overcome by additional dosing. These results demonstrate the potential of early infliximab TDM to guide decision-making in ASUC and for the first time provide an evidence base for intensified infliximab dosing in clinical practice.

We found that a lower serum infliximab level at day 3 was associated with a higher risk of induction failure at day 14 and colectomy at month 3. Day 3 serum infliximab levels were also numerically lower in day 7 infliximab non-responders although this did not reach statistical significance. This suggests that measuring serum infliximab on day 3 can help identify patients at high risk of treatment failure. Lower serum infliximab levels on day 3 ($\leq 57.9 \mu\text{g/mL}$) could indicate the need for early intensive redosing of infliximab, bearing in mind that these are early levels as opposed to trough levels.

This study supports observational data indicating infliximab clearance to be a major determinant of outcomes in ASUC,^{7,10,11} and for the first time demonstrates that increased clearance may potentially be overcome by more intensive dosing. In our study, we observed a significant association between early infliximab clearance based on serum infliximab levels between days 1 and 7 and day 7 infliximab nonresponse, day 14 infliximab failure, and month 3 colectomy. In a post-hoc subgroup analysis of the PREDICT-UC RCT, we also observed that an initial intensified 10 mg/kg infliximab dose was associated with improved outcomes compared with a standard 5 mg/kg dose in patients with high clearance. Additionally, we found that in patients with high clearance who do not respond to

a 5 mg/kg or 10 mg/kg first dose of infliximab, a 10 mg/kg second dose was superior to a 5 mg/kg dose in achieving response at day 14. Evaluation of response to infliximab based on CRP stratification yielded a numerically but not statistically significant difference between an initial 5 mg/kg and 10 mg/kg infliximab dose,⁶ indicating that infliximab clearance performs better than CRP in patient risk stratification.

We found that infliximab clearance is dynamic, highest initially and decreased over time. Although clearance at day 3 did not differ between initial responders and non-responders, clearance reduced more rapidly in responders. Similarly, fecal infliximab loss at day 3 was also found to be lower in responders. Infliximab responders tended to have a more favourable infliximab pharmacokinetic profile, characterized by a greater reduction in early clearance and lower stool infliximab losses compared with nonresponders. Our findings are consistent with 2 small retrospective studies in adults that observed an association between high clearance and week 14 clinical nonresponse,⁷ or high calculated baseline clearance and month 6 colectomy.¹¹ In a more recent study in the paediatric setting, Whaley et al¹⁰ also observed an association between high infliximab clearance and colectomy. Taken together, these results support the hypothesis that increased clearance contributes to infliximab failure. However, it remains uncertain whether the relationship between high clearance and infliximab failure is causal or if both are driven by underlying disease severity.

Whether fecal infliximab loss plays a role in infliximab failure has also been the subject of debate with inconsistent findings across 2 previous small studies.^{12,13} In our study, patients with lower fecal infliximab levels at day 1 had less severe endoscopic disease activity at baseline and had a higher likelihood of achieving Mayo endoscopic remission

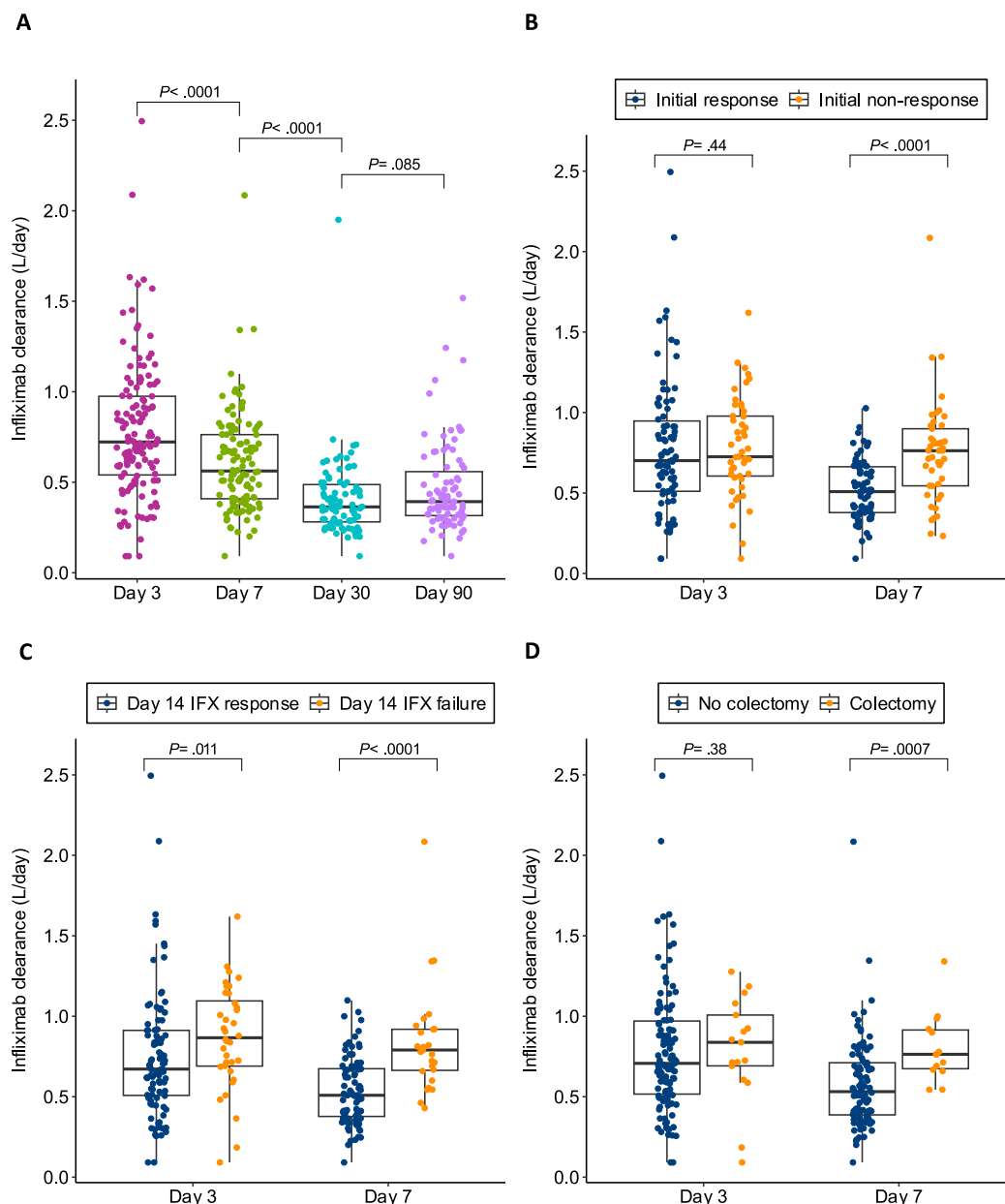


Figure 3. (A) IFX clearance over time; IFX clearance at day 3 and day 7 in (B) initial IFX responders and nonresponders, (C) day 14 IFX responders and failures, and (D) patients who avoided colectomy and those who required colectomy by month 3. IFX, infliximab.

and combined clinical and endoscopic remission at month 3. We also observed an association between lower day 3 fecal infliximab levels and improved day 7 response across the whole cohort. On subgroup analysis, this finding was only observed in the 5 mg/kg infliximab group, possibly due to saturation of fecal infliximab losses in the 10 mg/kg group. The correlation we observed between fecal infliximab level and clearance suggests that high fecal infliximab loss may at least in part contribute to high infliximab clearance. High fecal infliximab level may represent a surrogate marker of more severe endoscopic disease activity, which in turn drives high clearance. Regardless of the mechanism behind the correlation, our findings add weight

to the theory that disruption of mucosal integrity contributes to fecal infliximab loss.

An interesting finding from our study was that estimated early clearance was higher in patients who received an initial 10 mg/kg infliximab dose compared with those receiving a 5 mg/kg dose. This finding suggests “nonlinearity” of PK, which contradicts the standard two-compartment model with first order elimination that assumes that clearance is unrelated to dose. A potential explanation for this finding may relate to the higher fecal infliximab levels we observed in the 10 mg/kg group, suggesting that this group had higher fecal infliximab losses possibly due to the higher concentration gradient between

the colonic mucosa and the lumen in patients receiving a higher dose. Another potential explanation is that a higher degree of target-mediated drug deposition occurred in the mucosa of patients receiving a 10 mg/kg dose. Alternatively, this may relate to saturation of Fc receptor recycling, which is known to contribute to the duration of infliximab's half-life.

Our study also demonstrated the potential benefit of higher serum infliximab levels after successful induction of remission in ASUC. In patients who completed induction, a higher day 90 serum infliximab level was associated with a greater likelihood of Mayo endoscopic remission at month 3. This is in line with a study by Seow et al,¹⁴ which showed that, in a cohort of moderately severe to severe UC, a detectable trough serum infliximab level was linked to higher remission rates.

In the PREDICT-UC clinical trial from which our specimens were derived, we observed that IIS and AIS groups achieved earlier remission compared with the SIS group, however, the 3 groups had similar outcomes at month 3.⁶ This is likely attributable to the long interval (8–11 weeks) between the last infliximab induction dose to month 3 in the IIS and AIS group, as shown by the lower month 3 serum IFX levels in these 2 groups compared with the SIS group. Together, this suggests that in patients with ASUC, further infliximab doses after initial successful induction of remission (beyond the traditional 3 infliximab induction doses) may help consolidate outcomes at month 3, especially in patients who received an intensified or accelerated induction, or those with a high infliximab clearance.

Contrary to a study by Murthy et al,¹⁵ we found low rates of antibody to infliximab. This may be due to the fact that few of our patients had prior exposure to infliximab, hence they were less likely to develop immunogenicity. Alternatively, this may relate to differences in the antibody-to-infliximab assays used by Murthy et al¹⁵ compared with the drug-sensitive assay we used. Although drug-tolerant kits can more sensitively detect antibodies to infliximab, the clinical relevance of such antibodies especially in the presence of drug in the ASUC setting is unclear.

Our study has several strengths. Unlike other studies that have previously often comprised mixed cohorts of patients with both acute and chronically active moderate-to-severe UC,^{14,16} this is the largest prospective study to investigate infliximab levels and pharmacologic parameters in ASUC, and the first based on data from a multicenter RCT. To our knowledge, this study is the first to show a relationship between day 3 serum infliximab levels, the dynamic nature of infliximab clearance, and clinical outcomes. Additionally, it is novel in demonstrating for the first time that fecal infliximab levels correlate with the severity of endoscopic lesions and endoscopic outcomes, with endoscopic assessments conducted at both baseline and month 3. Because infliximab levels were only measured after the clinical study concluded, the results did not influence clinical management. Furthermore, the laboratory staff analyzing the specimens were blinded to clinical outcomes, which prevented measurement bias.

Although we identified early serum infliximab level and clearance as potential predictors of outcomes, whether

adapting the dosing strategy based on these parameters makes any difference to outcomes warrants prospective evaluation in a separate cohort. Although the recent TITRATE study did not demonstrate a benefit of dashboard-driven infliximab dosing in ASUC,¹⁷ the lack of benefit may have been due to an insufficient target infliximab level and dosing (target >28 µg/mL in the first 28 days) compared with our study, which showed that a day 3 serum infliximab level >57.9 µg/mL was associated with avoidance of colectomy. Our data suggest that a dashboard-driven approach that estimates clearance from early infliximab levels may improve outcomes.

Although our study demonstrated the potential benefit of infliximab drug monitoring in ASUC, clinical uptake may be limited by the current long turn-around time of infliximab level testing in many diagnostic laboratories. This may be improved by changes in the laboratory workflow in tertiary hospitals that routinely manage patients with ASUC or through point-of-care infliximab testing using rapid assays, although these still require validation in the ASUC setting where infliximab levels are much higher compared with those observed in outpatients. We also acknowledge that estimating infliximab clearance is currently challenging due to the limited availability of tools to calculate clearance readily in clinical practice. This may be overcome by development of calculators derived and validated from covariates in the ASUC setting.

A further limitation of our study is that we did not quantify total infliximab loss because measuring stool volume would have been impractical. Although we observed only a weak correlation between fecal infliximab levels and infliximab clearance, we hypothesize that total fecal infliximab loss may have shown a stronger correlation with infliximab clearance. Nevertheless, our study provides valuable insights into the mechanisms of infliximab failure and supports the hypothesis that infliximab clearance plays a role in determining outcomes in ASUC. Future research should explore other potential mechanisms of infliximab failure, including the role of tumor necrosis factor (TNF) burden and non-TNF-mediated immune pathways in infliximab nonresponse in ASUC.

Because our study was exploratory in nature, we conducted multiple hypothesis tests without formal adjustments for multiplicity. We acknowledge that this increases the risk of Type I errors, and, thus, the potential for spurious associations should be considered when interpreting the statistical significance of our findings.

In conclusion, early infliximab TDM has the potential to influence clinical decision-making and prognosticate outcomes in ASUC. A lower day 3 serum infliximab level predicted infliximab failure at day 14 and month 3 colectomy. A day 3 infliximab level threshold of >57.9 µg/mL had 83% specificity and 97% PPV for avoidance of colectomy. Infliximab clearance is dynamic, with high early clearance in the first week after infliximab associated with day 7 infliximab nonresponse, day 14 infliximab failure, and month 3 colectomy. Patients with low infliximab clearance respond equally to 5 mg/kg or 10 mg/kg dose strategy. A 10 mg/kg initial infliximab dose, or dose-intensified rescue

after initial nonresponse, may help overcome high clearance. Higher serum infliximab level at month 3 is associated with Mayo endoscopic remission and supports the administration of further infliximab doses between the end of induction and the month 3 mark to consolidate outcomes.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at <https://doi.org/10.1053/j.gastro.2025.07.020>.

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Conflicts of interest

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Data Availability

The data and protocol may be shared with researchers who submit a research proposal approved by the Austin Health Human Research Ethics Committee Board and at the discretion of the principal investigator (P.D.C.). If approval is granted, individual patient data supporting the reported results in this article will be shared in deidentified and anonymized datasets for the purpose of individual participant data meta-analysis. The data and protocol will be accessible for up to 24 months after publication. Proposals should be sent to the corresponding author and principal investigator at p.deacruz@unimelb.edu.au. Requestors will be required to sign a data access agreement. Proposals can be submitted within 24 months after publication.